

Part I

SUMMARY of Using Limits to Detect Asymptotes

Summary

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Infinite Limits

$\lim_{x \rightarrow a} f(x) = \infty$ means the values of $f(x)$ grow arbitrarily large as x approaches a .

$\lim_{x \rightarrow a} f(x) = -\infty$ means the values of $|f(x)|$ grow arbitrarily large as x approaches a with $f(x)$ negative.

Limits at Infinity

$\lim_{x \rightarrow \infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making x sufficiently large.

$\lim_{x \rightarrow -\infty} f(x) = L$ means the values of $f(x)$ becomes arbitrarily close to L by making $|x|$ sufficiently large with x negative.

Vertical and Horizontal Asymptotes:

A function f has a **vertical asymptote** at $x = a$ if at least one of the following conditions hold:

- $\lim_{x \rightarrow a} f(x) = \pm\infty$
- $\lim_{x \rightarrow a^+} f(x) = \pm\infty$
- $\lim_{x \rightarrow a^-} f(x) = \pm\infty$

A function f has a **horizontal asymptote** at $y = L$ if at least one of the following conditions hold:

- $\lim_{x \rightarrow \infty} f(x) = L$
- $\lim_{x \rightarrow -\infty} f(x) = L$

Both vertical asymptotes and horizontal asymptotes are written as the equation of a line. Vertical asymptotes are given as the equation of a vertical line, and horizontal asymptotes are given as the equation of a horizontal line.

Part II

Recitation Questions

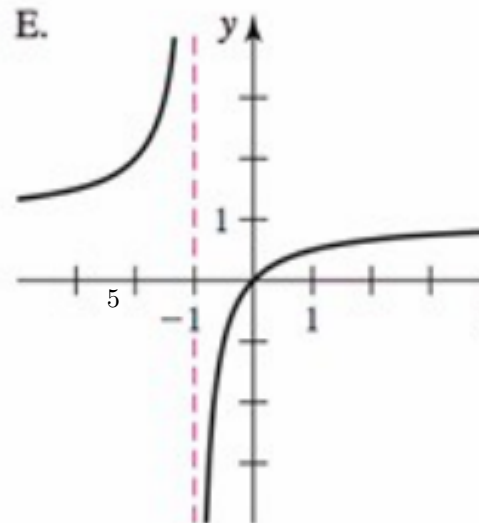
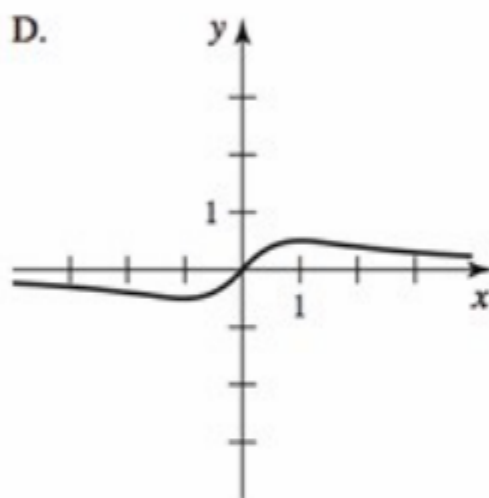
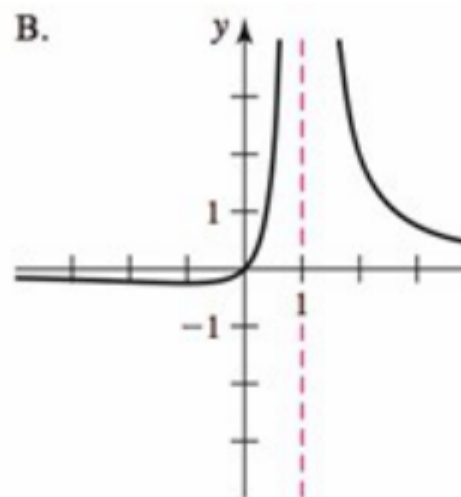
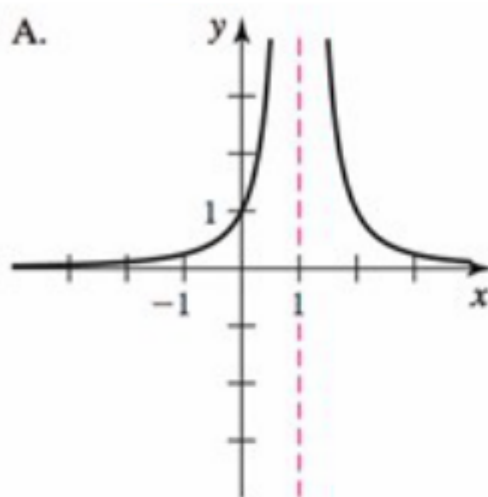
Problem 1

February 21, 2026

Problem 1

Problem 1 Without using a graphing utility, match each graph of functions in A-F with the algebraic representation of functions in a-f:

Problem 1



Problem 1

(a) The function f defined by $f(x) = \frac{x}{x^2 + 1}$.

(b) The function g defined by $g(x) = \frac{x}{x^2 - 1}$.

(c) The function h defined by $h(x) = \frac{1}{x^2 - 1}$.

(d) The function a defined by $a(x) = \frac{x}{(x - 1)^2}$.

(e) The function s defined by $s(x) = \frac{1}{(x - 1)^2}$.

(f) The function r defined by $r(x) = \frac{x}{x + 1}$.

Problem 2

February 21, 2026

Problem 2

Problem 2 Sketch a possible graph of a function g that satisfies the following conditions:

$$\text{Domain: } [-5, -2) \cup (-2, 3) \cup (3, 5)$$

$$\lim_{x \rightarrow -2} g(x) = 3$$

$$g(1) = 1$$

$$\lim_{x \rightarrow 1^+} g(x) = -\infty$$

$$\lim_{x \rightarrow 3} g(x) = \infty$$

$$\lim_{x \rightarrow 5^-} g(x) = \infty$$

$$\lim_{x \rightarrow 1^-} g(x) = 1$$

$$g(-5) = -1.8$$

Problem 3

February 21, 2026

Problem 3

Problem 3 Let f be a function given by $f(x) = \ln(1 + x)$.

- (a) Find the domain of f . Write your answer in interval notation.
- (b) Find the vertical asymptotes of f and **EXPLAIN** and justify your answer.
- (c) Sketch a graph of f

Problem 4

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Problem 4

Problem 4 Let $f(x) = \frac{\ln(x)}{x-2}$.

(a) Evaluate the limit.

$$\lim_{x \rightarrow 2^-} \frac{\ln(x)}{x-2}$$

(b) Find the vertical asymptotes of f . **EXPLAIN** and justify your answer.

Problem 5

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Problem 5

Problem 5 Select the meaning of $\lim_{x \rightarrow \infty} f(x) = 6$. Support your explanation graphically.

- (a) As x becomes arbitrarily negatively large, $f(x)$ approaches 6.
- (b) As x becomes arbitrarily positively large, $f(x)$ approaches 6.
- (c) As x approaches 6, $f(x)$ becomes arbitrarily negatively large.
- (d) As x approaches 6, $f(x)$ becomes arbitrarily positively large.

Problem 6

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Problem 6

Problem 6 *Evaluate the following limits.*

(a) $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x^9 + 5}}{3x^3 + \sqrt{4x^6 + 1}}$

(b) $\lim_{x \rightarrow \infty} \frac{\sin(9x)}{5x}$

Problem 7

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Problem 7

Problem 7 The function f is defined by $f(x) = \frac{6e^x + 1}{3e^x + 5}$.

- (a) Find all vertical asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.

Problem 8

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Problem 8

Problem 8 The function f is defined by $f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$.

- (a) Find all vertical asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes of f . **EXPLAIN** and justify your answer by using appropriate limits..

Problem 9

February 21, 2026

Problem 9 A piecewise defined function f is given by

$$f(x) = \begin{cases} \frac{2x-3}{x-2} & \text{if } x < 2 \\ \frac{x^2-5x+6}{x^2-4} & \text{if } x > 2 \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Problem 10

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Problem 10 For the piecewise function f defined by

$$f(x) = \begin{cases} \sin(x) & \text{if } x \leq 0 \\ \frac{x^2}{x^2 - 4} & \text{if } 0 < x < 2 \\ \frac{x^2}{x^2 + 4} & \text{if } 2 \leq x \end{cases}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Problem 11

February 21, 2026

Problem 11 For the function g defined by

$$g(t) = \frac{t^2 + 7t + 11}{t - 3}$$

- (a) Find all vertical asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.
- (b) Find all horizontal asymptotes. **EXPLAIN** and justify your answer by using appropriate limits.

Problem 12

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Problem 12

Problem 12 Sketch a possible graph of a function that satisfies all of the given properties. (You do not need to find a formula for the function.)

$$\begin{array}{lll}
 \lim_{x \rightarrow -2^-} f(x) = \infty & f(-2) = -5 & f(1) = 2 \\
 \lim_{x \rightarrow \infty} f(x) = -\infty & \lim_{x \rightarrow -\infty} f(x) = 4 & \lim_{x \rightarrow 5} f(x) = \infty \\
 \lim_{x \rightarrow 3} f(x) = 3 & f(3) = 1 & \\
 \lim_{x \rightarrow -2^+} f(x) = -5 & f(4) \text{ is undefined} & \lim_{x \rightarrow 4} f(x) = 3
 \end{array}$$